Cardiovascular System (CARD) · NPLEX-P1-CARD-11

CV-11 Hemodynamics and Blood Flow Regulation

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-11-01-C01-BD1

Brain dump 1 · foundational retrieval: Write the flow equations and Poiseuille's law from memory

A competent response contains

· Q = ΔP / R, the hydraulic form of Ohm's law; rearranged, R = ΔP / Q.

· Resistance is calculated rather than measured because only the pressure difference and the flow can be measured directly.

· For the whole systemic circulation the driving pressure is mean arterial pressure minus right atrial pressure, which is near 0 mm Hg, the flow is cardiac output, and the resistance is total peripheral resistance, so MAP = CO x TPR.

· Resting values: MAP about 93 mm Hg, cardiac output about 5 L/min, total peripheral resistance about 18 to 19 mm Hg per L/min.

· Pressure rises only if cardiac output or total peripheral resistance rises, and a fall in one must be offset by a rise in the other to hold pressure steady.

· Poiseuille's law: $R = 8\eta L / (\pi r^4)$, where η is viscosity, L is vessel length and r is internal radius.

· Resistance is directly proportional to viscosity and to length, and inversely proportional to the fourth power of the radius.

· The fourth power arises because laminar flow has a parabolic velocity profile: a wider vessel has both a larger cross-sectional area, which goes with r squared, and a larger central core of fluid far from the slowing effect of the wall, so velocity in that core rises as well.

· Halving the radius raises resistance 16-fold; widening the radius by about 19% halves resistance and doubles flow, because 1.19 to the fourth power is about 2.

· Length is fixed in the adult, but it explains why long diffuse disease resists more than a short lesion of the same caliber.

· Assumptions: a rigid tube, laminar flow and a Newtonian fluid. Passive vessels distend as pressure rises, so flow rises more than in proportion; autoregulating vessels constrict, so flow stays nearly constant; turbulent flow needs a larger gradient for the same flow. Use the law for direction and relative size of change.

Distinctions that matter

· Resistance and flow move in opposite directions for the same change in radius, so state which one the question asks for before doing the arithmetic.

· Viscosity, length and the pressure gradient change flow only in direct proportion, while radius changes it by the fourth power.

Common misconceptions

· Writing that a doubled pressure gradient halves resistance. In a rigid tube resistance is unchanged and flow doubles; it is the vessel wall, not the gradient, that sets resistance.

· Treating the fourth power as a property of area alone. Area accounts for only two of the four powers, and the velocity profile accounts for the other two.

Where it is taught: Section C01, opening paragraph and the paragraph on Poiseuille's law of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C01-BD2

Brain dump 2 · integration and application: Work out what single changes do to resistance and flow

A competent response contains

· Radius halved: resistance increases 16-fold and flow falls to one sixteenth. Example, arteriolar constriction by sympathetic alpha-1 stimulation or a severe focal stenosis.

· Radius doubled: resistance falls to one sixteenth and flow increases 16-fold. Example, metabolic dilation of muscle arterioles during exercise.

· Radius widened about 19%: resistance halves and flow doubles. Example, a small dilation of a coronary arteriole when myocardial oxygen demand rises.

· Viscosity doubled: resistance doubles and flow halves. Example, severe polycythemia or a hyperviscosity state; anemia does the reverse.

· Length doubled: resistance doubles and flow halves. Example, long diffuse arterial disease compared with a short focal lesion.

· At a hematocrit of 68% viscosity is raised two to three times, so resistance rises in proportion and flow falls.

· Oxygen delivery is flow times oxygen content, so when flow falls by more than content rises, delivery drops; sluggish flow also favors thrombosis.

· Other causes of raised viscosity: large plasma proteins such as an IgM paraprotein in Waldenström macroglobulinemia or a high fibrinogen, rigid sickled cells, and extreme leukocytosis.

· Blood is non-Newtonian: at low flow red cells form rouleaux and viscosity rises, which matters in low-flow states and small veins.

· In vessels below about 0.3 mm a cell-poor plasma layer at the wall lowers apparent viscosity, the Fåhræus-Lindqvist effect, which matters in arterioles and capillaries.

· Whole blood at a hematocrit of 40 to 45% is roughly 3 to 4 times as viscous as water, and hematocrit is the main determinant.

Distinctions that matter

· Hematocrit sets viscosity, which changes resistance in direct proportion, while radius changes it by the fourth power, so a modest vasoconstriction outweighs a large change in hematocrit.

· Rouleaux formation raises apparent viscosity in slow large-vessel flow, while the cell-poor wall layer lowers it in the smallest vessels, so the two effects belong to different parts of the circuit.

Common misconceptions

· Assuming more red cells always means more oxygen reaching the tissue. Once viscosity rises steeply, the loss of flow costs more than the extra content gains.

· Treating blood as a Newtonian fluid at all flow rates. Its apparent viscosity depends on shear rate and on vessel caliber.

Where it is taught: Section C01, the table on predicting a single change and the paragraph on viscosity of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C01-PR01

Prediction exercise: Predict the effect of a focal arterial stenosis

A competent response contains

· Resistance of the narrowed segment rises 16-fold, because resistance varies with the inverse fourth power of the radius.

· Flow through the segment falls toward one sixteenth of its former value if the gradient does not change, from $Q = \Delta P / R$.

· Area falls to one quarter, because area goes with the radius squared, so velocity inside the stenosis rises about 4-fold, from $v = Q / A$ and the continuity principle.

· Reynolds number is $\rho v D / \eta$, so a 4-fold rise in velocity with a halved diameter roughly doubles it, and the jet breaks into eddies as it enters the wider vessel beyond.

· A bruit is heard, because turbulent eddies vibrate the vessel wall; the same disturbance can be felt as a thrill.

· As the stenosis becomes critical, flow, velocity and the Reynolds number all fall, so the bruit softens or disappears, which is why a quiet carotid does not exclude severe stenosis.

· A long diffuse diseased segment resists more than a short lesion of the same caliber, because resistance is directly proportional to length.

· Turbulence also wastes energy, so a larger pressure gradient is needed for the same flow, which adds to the effective resistance of the segment.

Distinctions that matter

· Flow through the whole limb falls while velocity inside the narrowed segment rises, because flow and velocity are linked through area rather than being the same quantity.

· A loud bruit reports a high-velocity jet, not the severity of the narrowing, so loudness and severity part company once the lesion becomes critical.

Common misconceptions

· Reading a louder bruit as a worse stenosis in every case. A critical lesion can be silent because the jet has lost its velocity.

· Expecting velocity to fall inside a narrowing because flow has fallen. The same flow must cross a much smaller area, so velocity rises.

Where it is taught: Section C01, Poiseuille paragraph and table; Section C03, paragraphs on continuity and on laminar and turbulent flow of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C01-Q01

Board-style question: Arteriolar dilation and flow

Correct answer: C. Flow approximately doubles

Why it is right: Flow at a constant pressure gradient varies with the fourth power of the radius, and 1.19 to the fourth power is about 2, which is why a local dilation of roughly 19% can double flow to one muscle without any change in arterial pressure or in flow elsewhere.

A is wrong: A 19% rise in flow would follow if flow varied in direct proportion to radius, which applies to neither resistance nor flow.

B is wrong: A 50% rise follows from scaling flow with cross-sectional area, which accounts for only two of the four powers of the radius.

D is wrong: A 16-fold rise follows from doubling the radius, not from widening it by about 19%.

A competent response contains

· Resistance varies with the inverse fourth power of the radius, so flow at a fixed pressure gradient varies with the fourth power of the radius.

· A radius 1.19 times normal gives 1.19 to the fourth power, which is about 2, so flow approximately doubles.

Distinctions that matter

· A small change in radius produces a large change in flow, which is why arterioles rather than arterial pressure set regional flow.

Common misconceptions

· Scaling flow in direct proportion to radius, or in proportion to area, both of which understate the change.

Where it is taught: Section C01 of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C02-BD1

Brain dump 1 · foundational retrieval: Series and parallel resistance, and the pressure profile

A competent response contains

· In series: R total = R1 + R2 + ..., the same flow passes through every segment, and the segment with the highest resistance takes the largest pressure drop because $\Delta P = Q \times R$.

· In parallel: 1 / R total = 1 / R1 + 1 / R2 + ..., every branch sees the same pressure difference, flows add, and conductances add.

· Total parallel resistance is lower than the smallest single branch, and n identical branches of resistance R give R / n.

· Worked in series: 10 + 20 + 20 = 50 units. Worked in parallel: 1/10 + 1/20 + 1/20 = 0.2, so the total is 5 units.

· Aorta and large arteries: about 95 to 100 mm Hg and pulsatile, small share of resistance, aorta about 2.5 cm^2, and all systemic arteries hold about 13% of blood volume.

· Small arteries and arterioles: pressure falls from about 85 to 90 mm Hg down to about 30 to 35 mm Hg, the largest share of resistance with arterioles dominating, arterioles about 40 cm^2, about 7% of volume together with the capillaries.

· Capillaries: about 30 to 35 mm Hg at the arterial end and 10 to 15 mm Hg at the venous end, moderate resistance because so many lie in parallel, about 2,500 cm^2.

· Venules and veins: about 10 to 15 mm Hg falling to near 0 at the right atrium, small share of resistance, venules about 250 cm² and the venae cavae about 8 cm², and systemic veins hold about 64% of blood volume.

· Arterioles hold the largest adjustable share because they are relatively few, they are narrow, and they carry thick smooth muscle relative to their lumen, so their radius can be changed over a wide range.

· Capillaries are individually narrower than arterioles but there are billions in parallel, which gives only a moderate combined resistance, and they have no smooth muscle, so their caliber cannot be adjusted.

· Terminal arterioles and precapillary sphincters decide how many capillaries are open, and the steep arteriolar pressure drop also damps the arterial pulse so capillary flow is nearly steady.

· The heart holds about 7% and the lungs about 9% of blood volume.

Distinctions that matter

· Resistances add in series while conductances add in parallel, so adding a branch raises total resistance in one arrangement and lowers it in the other.

· Having the narrowest lumen and having the largest share of resistance are different claims: capillaries are narrower, arterioles resist more, and the number of vessels in parallel is what reconciles the two.

Common misconceptions

· Adding parallel resistances directly. Total parallel resistance must come out lower than the smallest branch, which is a quick check on the arithmetic.

· Naming the capillaries as the main site of resistance because they are the narrowest vessels. Their enormous parallel number keeps their combined resistance moderate.

Where it is taught: Section C02, opening paragraph, the paragraph on why arterioles set total peripheral resistance, and the table on pressure, area, velocity and volume of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C02-BD2

Brain dump 2 · integration and application: Adding and removing parallel paths, and beds placed in series

A competent response contains

· An arteriovenous fistula is a low-resistance path added in parallel, so total peripheral resistance falls.

· Mean arterial pressure would fall if cardiac output did not change, so baroreceptors raise sympathetic output and cardiac output rises to defend pressure.

· Rapid diastolic runoff through the fistula lowers diastolic pressure and widens pulse pressure.

· A very large fistula keeps cardiac output high over time and can cause high-output heart failure.

· A mid-thigh amputation removes a parallel branch, so total peripheral resistance rises. With branches of 10, 20 and 20 units the total is 5 units, and removing the 10-unit branch raises it to 10 units.

· Sequence when a large bed dilates: one large bed dilates, such as exercising muscle; total peripheral resistance falls; mean arterial pressure would fall if cardiac output were unchanged; baroreceptors raise sympathetic output; cardiac output rises while gut, kidney and inactive muscle constrict; mean arterial pressure is defended and flow shifts to the dilated bed.

· Because systemic organs branch from the aorta in parallel, each sees nearly the same arterial pressure and its flow equals mean arterial pressure divided by its own resistance, so one organ can raise its flow by dilating without starving the others.

· Beds in series: gut and spleen draining to the liver through the portal vein, glomerular capillaries to peritubular capillaries, and the hypothalamus to the anterior pituitary.

· In a series arrangement, raised downstream resistance raises upstream capillary pressure.

· In cirrhosis the fibrotic liver raises resistance downstream of the splanchnic capillaries, so splanchnic capillary pressure rises, which is portal hypertension.

· The pulmonary and systemic circuits are in series with each other, so the two ventricles must have equal average outputs.

Distinctions that matter

· Adding a parallel path lowers total peripheral resistance while removing one raises it, and neither change by itself tells you what happens to pressure until the reflex change in cardiac output is included.

· Organs in parallel are protected from one another, while organs in series are not, which is why only the series beds develop upstream capillary hypertension.

Common misconceptions

· Expecting mean arterial pressure to fall and stay low after a fistula is created. The baroreceptor reflex raises cardiac output, so mean pressure is largely defended while diastolic pressure falls.

· Treating portal hypertension as a problem of blood volume alone. It follows from the series arrangement of splanchnic capillaries and the liver.

Where it is taught: Section C02, opening paragraph, final paragraph, the flow figure on opening a parallel bed and the stop and predict box of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C03-BD1

Brain dump 1 · foundational retrieval: Velocity against total cross-sectional area

A competent response contains

· v = Q / A, so velocity is flow divided by cross-sectional area.

· The same cardiac output passes through every level of the series circuit, so velocity at each level varies inversely with the total cross-sectional area at that level.

· Mean velocity is about 33 cm/s in the aorta and about 0.3 mm/s in the capillaries.

· The combined capillary area is about 1,000 times larger than the aortic area, which is why capillary velocity is lowest.

· On the sketch, total cross-sectional area rises to a peak at the capillaries and falls again toward the venae cavae, while velocity is the mirror image of that curve.

· A red cell spends about 1 to 3 seconds in a systemic capillary, and that transit time allows diffusion of gases, nutrients and wastes.

· Velocity rises again in the veins because total area falls, but it stays below aortic values because the venae cavae together have a larger area than the aorta, about 8 cm^2 against about 2.5 cm^2.

· Locally the continuity principle means a narrowed segment carries the same flow through a smaller area, so velocity rises there.

Distinctions that matter

· Flow is volume per time and velocity is distance per time, and area is what links them, so a low capillary velocity does not mean low capillary flow.

· The whole-circuit relationship between velocity and total area is a different use of the same equation from the local rise in velocity through a stenosis.

Common misconceptions

· Saying capillary velocity is low because each capillary is narrow. A narrow tube carrying the same flow would have a higher velocity; it is the number of capillaries in parallel that lowers it.

· Expecting venous velocity to match aortic velocity because the same cardiac output returns. Total venous area is larger, so velocity stays lower.

Where it is taught: Section C03, opening paragraph and the paragraph on the continuity principle of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C03-BD2

Brain dump 2 · integration and application: Turbulence at the bedside

A competent response contains

· Re = ρvD / η, where ρ is density, v is velocity, D is diameter and η is viscosity. Rising velocity, rising diameter and falling viscosity all raise it.

· Density barely varies in the body, so velocity, diameter and viscosity are the clinical variables.

· Turbulence appears above a Reynolds number of about 2,000 in a straight smooth tube, and at lower values at branches, curves and abrupt changes in caliber.

· Turbulent flow forms eddies that vibrate the vessel wall, which is what makes a murmur, a bruit or a palpable thrill, and the eddies waste energy, so a larger pressure gradient is needed for the same flow.

· Anemia sequence: hemoglobin and hematocrit fall; viscosity falls and resistance falls; low oxygen content dilates arterioles so total peripheral resistance falls further; cardiac output and stroke volume rise; ejection velocity through normal valves rises; higher velocity and lower viscosity raise the Reynolds number; a soft systolic flow murmur appears across normal valves.

· Anemia acts on velocity and viscosity at once, which makes it the classic cause of a flow murmur, while polycythemia raises viscosity and tends to abolish flow murmurs.

· A carotid stenosis is audible because the jet is fast and the caliber changes abruptly; a critical stenosis can be silent because flow, velocity and the Reynolds number have all fallen.

· An arteriovenous fistula connects a high-pressure to a low-pressure vessel, giving a very high velocity and a continuous bruit.

· Just below systolic pressure the cuff leaves a narrowed segment through which blood spurts at each peak, and that turbulence is the first Korotkoff sound; once cuff pressure falls below diastolic pressure flow is laminar throughout and the sound disappears, which is phase V.

· Other causes of raised velocity and flow murmurs: pregnancy, fever, hyperthyroidism and exercise. A dilated aortic root raises the Reynolds number through diameter.

· Shear stress is low and oscillating at branch points and bifurcation walls, which reduces endothelial nitric oxide and favors atherosclerosis, while steady laminar high shear keeps the endothelium quiescent and antithrombotic.

· Jets beyond a stenotic valve can cause post-stenotic dilation.

· The simplified Bernoulli relation estimates the gradient in mm Hg as about 4 times the peak velocity in m/s squared.

Distinctions that matter

· Low viscosity raises the Reynolds number while high viscosity lowers it, so anemia favors turbulence for two reasons at once and polycythemia opposes it.

· Velocity rather than flow determines whether turbulence occurs, which is why a critically narrowed vessel with very low flow can be silent.

Common misconceptions

· Taking a quiet carotid as evidence against severe stenosis. Loss of the bruit can mean the lesion has become critical.

· Treating every systolic murmur as valve disease. Anemia, fever, pregnancy, hyperthyroidism and exercise produce murmurs across structurally normal valves.

Where it is taught: Section C03, paragraph on laminar and turbulent flow, the table on factors favoring turbulence, the flow figure on anemia and the final paragraph on Korotkoff sounds and shear stress of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C03-RT01

Retrieval exercise: Rebuild the sequence from anemia to a flow murmur

A competent response contains

· Step 1: hemoglobin and hematocrit fall.

· Step 2: viscosity falls and resistance falls. Falling viscosity lowers resistance, which raises flow, and it also raises the Reynolds number directly, so the two effects reach the murmur through different variables.

· Step 3: low oxygen content dilates arterioles, so total peripheral resistance falls further. The dilating stimulus is the low oxygen content of the blood reaching the tissue.

· Step 4: cardiac output and stroke volume rise, and stroke volume is the component that rises most.

· Step 5: ejection velocity through normal valves rises.

· Step 6: higher velocity and lower viscosity raise the Reynolds number.

· Step 7: a soft systolic flow murmur is heard across normal valves.

· The same reasoning explains flow murmurs in pregnancy, fever and hyperthyroidism.

Distinctions that matter

· Raised velocity and lowered viscosity are two separate routes into the Reynolds number, and anemia supplies both, which is why its murmur is so reliable.

· The murmur arises across structurally normal valves, so it reports the state of the blood and the output rather than the valve.

Common misconceptions

· Expecting lower viscosity to quiet the circulation because resistance has fallen. Lower viscosity raises the Reynolds number and the higher output raises velocity, so turbulence becomes more likely.

· Putting the arteriolar dilation before the fall in viscosity. The viscosity change follows directly from the lower hematocrit, and the dilation is a response to low oxygen content.

Where it is taught: Section C03, the flow figure on anemia to a flow murmur and the table on factors favoring turbulence of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C04-BD1

Brain dump 1 · foundational retrieval: Compliance, the arterial reservoir and pulse pressure

A competent response contains

· C = ΔV / ΔP, and its reciprocal is stiffness, also called elastance.

· Systemic veins are about 24 times as compliant as the corresponding arteries, roughly 8 times as distensible with about 3 times the volume, which is why they hold most of the blood.

· During ejection the elastin-rich large arteries stretch and store part of the stroke volume; during diastole they recoil and push that stored volume onward.

· That reservoir function turns pulsatile output into near-continuous capillary flow, keeps diastolic pressure from falling to zero, and lowers the peak pressure the ventricle must generate.

· Pulse pressure is systolic minus diastolic pressure, normally about 40 mm Hg, and is roughly proportional to stroke volume divided by arterial compliance.

· Diastolic pressure depends on how fast the stored volume runs off, governed by the time constant R x C.

· Low resistance, low compliance or a long diastole each let diastolic pressure fall lower.

· Mean arterial pressure is about diastolic pressure plus one third of the pulse pressure at resting heart rates.

· That estimate loses accuracy at fast heart rates, because systole then takes a larger share of each cycle.

Distinctions that matter

· Compliance and resistance are separate properties of the arterial system: resistance with cardiac output sets mean pressure, while compliance with stroke volume sets pulse pressure.

· Distensibility is a fractional change in volume while compliance is an absolute change per unit pressure, which is why the venous figures of about 8 and about 24 differ.

Common misconceptions

· Treating the arterial reservoir as a store of blood volume. The veins hold the volume; the arteries store part of each stroke volume for a fraction of a second and return it in diastole.

· Assuming low compliance means low mean pressure. Stiffening changes the shape of the pressure wave while mean pressure stays set by cardiac output and total peripheral resistance.

Where it is taught: Section C04, first two paragraphs of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C04-BD2

Brain dump 2 · integration and application: Predict the pressure pattern in a stiff aorta

A competent response contains

· Pulse pressure is roughly stroke volume divided by compliance, so halving compliance with the same 70 mL stroke volume roughly doubles pulse pressure, from 40 mm Hg to about 80 mm Hg.

· The runoff time constant R x C is also halved, so pressure falls faster and further before the next beat, which lowers diastolic pressure.

· Mean arterial pressure is set by cardiac output and total peripheral resistance, neither of which has changed, so it stays about 93 mm Hg.

· A pressure of about 120/80 mm Hg in the younger person becomes something like 150/65 mm Hg in the older one, so systolic pressure rises, diastolic pressure falls and mean pressure barely moves.

· Wall changes with age: elastin fragments, collagen increases and is cross-linked by advanced glycation end products, and the media may calcify.

· Hypertension, diabetes and chronic kidney disease accelerate those changes.

· Faster pulse wave velocity returns the reflected wave in late systole, so it augments the systolic peak instead of supporting diastolic pressure.

· Isolated systolic hypertension is the result and is the most common form in older adults, with diastolic pressure levelling off or falling after about age 50 to 60. The 2017 ACC/AHA guideline uses a systolic of 130 mm Hg or higher with a diastolic below 80 mm Hg; other guidelines use 140 and 90 mm Hg.

· Consequences: higher afterload drives concentric left ventricular hypertrophy, and lower diastolic pressure reduces coronary perfusion, which for the left ventricle occurs mainly in diastole, so the subendocardium is most exposed.

· The wider pulsatile pressure reaches the low-resistance brain and kidney, contributing to lacunar infarcts, white matter disease and declining renal function, and stiffness itself rises with stretch, so hypertension and stiffening reinforce each other.

· Widens: aortic regurgitation, through a large stroke volume plus diastolic backflow; hyperthyroidism, through a larger stroke volume with lower total peripheral resistance; arteriovenous fistula, through low resistance and rapid diastolic runoff; exercise, through a larger and faster stroke volume, mainly raising systolic pressure.

· Narrows: hemorrhage, aortic stenosis and cardiac tamponade, all through a small stroke volume, with reflex constriction holding diastolic pressure up in hemorrhage.

Distinctions that matter

· A wide pulse pressure from stiffening comes with a normal mean pressure, while a wide pulse pressure from a large stroke volume or a low resistance usually comes with other signs of a high output state.

· Systolic and diastolic pressure can move in opposite directions, which is why mean arterial pressure can look unchanged in a patient whose pressure wave has altered considerably.

Common misconceptions

· Explaining isolated systolic hypertension by a raised cardiac output. Stroke volume and total peripheral resistance are unchanged; it is compliance that has fallen.

· Treating a low diastolic pressure in an older adult as reassuring. It reduces the pressure driving left ventricular coronary flow during diastole.

Where it is taught: Section C04, the table on pulse pressure, the paragraph on aging and isolated systolic hypertension, the flow figure and the worked example of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the capillary and its Starling forces

A competent response contains

· Jv = Kf x [(Pc - Pi) - σ(πc - πi)], where Kf is the filtration coefficient, Pc is capillary hydrostatic pressure, Pi is interstitial hydrostatic pressure, σ is the reflection coefficient, πc is plasma oncotic pressure and πi is interstitial oncotic pressure.

· Teaching values: Pc about 30 mm Hg at the arterial end and 10 mm Hg at the venous end, Pi about -3 mm Hg, πc about 28 mm Hg, πi about 8 mm Hg.

· Pc pushes fluid out, a negative Pi adds to filtration, πc pulls fluid in and πi pulls it out.

· Arterial end: (30 + 3) - (28 - 8) = 33 - 20 = +13 mm Hg, so fluid filters out.

· Venous end: (10 + 3) - 20 = -7 mm Hg, so fluid is reabsorbed.

· Filtration slightly exceeds reabsorption, and the lymphatics return about 2 to 3 L per day along with the protein that has leaked out.

· Albumin supplies about 80% of plasma oncotic pressure, because it is the most abundant plasma protein and is poorly permeant.

· The reflection coefficient is near 1 in continuous capillaries and low in hepatic sinusoids; the filtration coefficient is high in glomeruli and low in the brain.

· In the revised glycocalyx model most capillaries filter along their whole length, and both models predict edema identically.

· Small solutes cross the capillary wall freely, so water movement is set by hydrostatic pressure and by the oncotic pressure of poorly permeant proteins.

Distinctions that matter

· Hydrostatic and oncotic pressures are not two versions of the same force: one is a mechanical push and the other is an osmotic pull created by proteins the wall holds back.

· Kf reports how leaky the wall is to water and σ reports how well it holds back protein, so a wall can change in one without the other.

Common misconceptions

· Taking a negative interstitial pressure as a sign that something has gone wrong. Loose tissue normally sits at about -3 mm Hg, and that slight suction adds to filtration.

· Leaving interstitial oncotic pressure out of the arithmetic. At about 8 mm Hg it cancels nearly a third of the plasma oncotic pull.

Where it is taught: Section C05, first two paragraphs of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-BD2

Brain dump 2 · integration and application: Work out the edema in four patients

A competent response contains

· Patient 1, nephrotic syndrome: πc falls, the fluid is a transudate, the swelling pits, and it is generalized, appearing first where tissue pressure is low, such as periorbital and scrotal tissue, and in dependent areas.

· Patient 1 arithmetic with πc at 16 mm Hg: the arterial end becomes 33 - (16 - 8) = 33 - 8 = +25 mm Hg, up from +13 mm Hg; the venous end becomes 13 - 8 = +5 mm Hg, so a site that formerly reabsorbed now filters.

· Patient 1 is generalized because the fall in plasma oncotic pressure affects every capillary in the body at once.

· Patient 2, right heart failure: Pc rises because venous pressure is raised, the fluid is a transudate, the swelling pits, and it appears in dependent areas with ascites and hepatic congestion. Reduced effective arterial volume activates the renin-angiotensin-aldosterone system and ADH, whose sodium and water retention raises venous pressure further.

· Patient 3, deep burn: permeability rises, so Kf rises, σ falls and πi rises; the fluid is a protein-rich exudate and the swelling is localized to the injured forearm.

· Patient 4, nodes removed and radiated: lymphatic removal fails and πi rises; capillary forces are normal at first; the fluid is protein rich and the swelling is localized to the arm.

· Patient 4 stops pitting because chronic protein-rich fluid drives fibrosis and fat deposition, and fibrosed tissue no longer indents.

· Safety factors: interstitial pressure rises quickly and opposes further filtration, lymph flow increases many-fold, and lymph washes interstitial protein out, which lowers πi.

· Guyton and Hall estimate those factors allow Pc to rise about 17 mm Hg before significant edema appears.

· Above an interstitial pressure of about 0 mm Hg the interstitium becomes highly compliant, so large volumes accumulate for little further rise in pressure.

· Transudates have a specific gravity below about 1.012 and exudates above about 1.020, so the fluid points back to the mechanism.

· Protein washout blunts the early change in nephrotic syndrome, which is why plasma oncotic pressure must fall substantially before edema appears.

Distinctions that matter

· Low plasma oncotic pressure produces generalized edema while raised capillary pressure and lymphatic failure produce edema where the pressure problem is, so distribution points at the term.

· Pitting reports fluid that moves under pressure, so a transudate keeps pitting while protein-rich fibrosed lymphedema stops.

Common misconceptions

· Expecting lymphedema to show abnormal Starling pressures at the capillary. The forces start normal, and the failure is in removal.

· Assuming any fall in albumin causes edema. The safety factors, protein washout in particular, must be overcome first.

Where it is taught: Section C05, the paragraph on mechanisms of edema, the edema table, the safety factor paragraph, the worked example on hypoalbuminemia and the stop and predict box of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-CP01

Comparison table: The four mechanisms of edema side by side

A competent response contains

· Raised capillary hydrostatic pressure: Pc rises; heart failure, deep vein thrombosis, portal hypertension, renal volume overload, prolonged standing and dihydropyridine arteriolar dilation; transudate; low protein content with a specific gravity below about 1.012; pits and keeps pitting; localized to the obstructed or dependent territory.

· Low plasma oncotic pressure: πc falls; nephrotic syndrome, cirrhosis, protein malnutrition such as kwashiorkor, and protein-losing enteropathy; transudate; low protein content with a specific gravity below about 1.012; pits; generalized, appearing first periorbitally and in dependent areas.

· Increased permeability: Kf rises, σ falls and πi rises; inflammation, burns, sepsis, ARDS and angioedema; exudate; protein rich with a specific gravity above about 1.020; pits early; localized in inflammation and widespread in sepsis.

· Lymphatic obstruction: removal fails and πi rises; node dissection or radiation, tumor, and filariasis from Wuchereria bancrofti; protein rich; specific gravity in the exudate range; pits at first and stops pitting once the tissue fibroses; localized to the drainage territory of the blocked lymphatics.

· Edema appears whenever filtration exceeds lymphatic removal, so each mechanism is a different way of crossing that same threshold.

Distinctions that matter

· Protein content separates the mechanisms that leave the wall intact from the two that do not: raised Pc and low πc give a transudate, while increased permeability and lymphatic obstruction give protein-rich fluid.

· Increased permeability lets plasma protein across an abnormal wall, while lymphatic obstruction traps protein that crossed a normal one.

Common misconceptions

· Treating all edema as pitting. Fibrosed chronic lymphedema does not pit.

· Assuming cirrhosis acts through one term. Portal hypertension raises splanchnic Pc while a low albumin lowers πc, which is why ascites forms so readily.

Where it is taught: Section C05, the paragraph on mechanisms of edema and the edema table of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C05-CA01

Clinical application: Abdominal swelling and ankle edema in a man with liver disease

A competent response contains

· Recognize: capillary hydrostatic pressure is raised in the splanchnic bed by portal hypertension, and plasma oncotic pressure is lowered body-wide by the albumin of 2.1 g/dL, so Pc is up where the liver drains and πc is down everywhere.

· Explain: the gut and splenic capillaries drain through the portal vein into the hepatic sinusoids, so they lie in series with the liver. A fibrotic cirrhotic liver raises resistance downstream of those capillaries, and raised downstream resistance raises upstream capillary pressure, which is portal hypertension; the dilated abdominal wall veins are collateral flow around that obstruction. The failing liver also synthesizes less albumin, and since albumin supplies about 80% of plasma oncotic pressure, πc falls and net filtration rises in every capillary, including those of the ankles.

· Differentiate: right heart failure would raise the jugular venous pressure and congest the liver from the venous side, and the jugular venous pressure here is not raised and the lungs are clear. Lymphatic obstruction would give protein-rich fluid in one drainage territory that eventually stops pitting, whereas this fluid has a specific gravity of 1.009, which is a transudate, and the ankle edema pits and is bilateral.

· Investigate: serum albumin, which is 2.1 g/dL against a reference range of 3.5 to 5.0 g/dL; the specific gravity of the ascitic fluid, which is 1.009 and therefore below the transudate threshold of about 1.012; jugular venous pressure, which is not raised; shifting dullness and dilated abdominal wall veins, which indicate free fluid and portal collateral flow; and the pitting character of the ankle edema with a slow refill.

· Decide: the fluid collects in the abdomen first because the raised capillary hydrostatic pressure is confined to the splanchnic bed drained by the portal vein, while the low oncotic pressure acts everywhere and adds the dependent ankle edema. The specific gravity of 1.009 marks a transudate, so the capillary wall is intact and protein has not leaked across it.

· Justify: the abdominal fluid reflects a high Pc term in the splanchnic capillaries added to a low πc term, while the ankle edema reflects the low πc term with the help of gravity on Pc. The safety factors are overwhelmed because two terms move at once and both move in the direction of filtration, the peritoneal cavity offers little opposing hydrostatic pressure, and once interstitial pressure passes about 0 mm Hg the tissue becomes highly compliant, so large volumes accumulate for little further rise in pressure.

Distinctions that matter

· Ascites with a normal jugular venous pressure points to a pressure problem upstream of the liver, while ascites with a raised jugular venous pressure points to one downstream of it.

· Two Starling terms moving together explain a distribution that neither term alone would give: abdominal fluid from the regional Pc and bilateral ankle edema from the body-wide πc.

Common misconceptions

· Attributing all of this patient's fluid to the low albumin. That would predict generalized edema with no preference for the abdomen, and it misses the series arrangement that raises splanchnic capillary pressure.

· Reading the dilated abdominal wall veins as a sign of heart failure. They are portal collaterals, which tells you where the raised pressure sits.

Where it is taught: Section C05, mechanisms of edema and the edema table, with the series bed and portal hypertension paragraph in C02 of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-BD1

Brain dump 1 · foundational retrieval: The four layers of control over arteriolar tone

A competent response contains

· Contraction: cytosolic Ca2+ rises, binds calmodulin, and the complex activates myosin light chain kinase, which phosphorylates the myosin light chains; relaxation follows a fall in Ca2+ or activation of myosin light chain phosphatase.

· Constrictors act through Gq receptors, where IP3 releases Ca2+, or by depolarization that opens L-type Ca2+ channels; dilators raise cGMP or cAMP, or open K+ channels and hyperpolarize the cell.

· Metabolic control: adenosine, CO2, H+, K+, lactate, phosphate and low PO2 accumulate when activity rises or flow falls short, and they dilate arterioles partly by opening K+ channels. Flow then washes them out, which is negative feedback. Active hyperemia is the rise in flow with raised tissue activity; reactive hyperemia is the surge after an occlusion, proportional to its duration.

· Myogenic control, the Bayliss response: stretch opens mechanosensitive cation channels, the cell depolarizes, L-type Ca2+ channels open, and the arteriole constricts; less stretch relaxes it.

· Endothelial control: shear stress and the agonists acetylcholine, bradykinin, histamine and substance P activate eNOS, and the nitric oxide pathway relaxes the smooth muscle, with the dilation spreading upstream to feeding arteries as flow-mediated dilation. Prostacyclin raises cAMP, hyperpolarizing factors matter in small arterioles, and endothelin-1 constricts through ETA receptors. With a damaged endothelium acetylcholine constricts instead, because it reaches muscarinic receptors on the smooth muscle with no nitric oxide to oppose it.

· Neural control: sympathetic norepinephrine acts on alpha-1 receptors, which are Gq coupled, and sets basal tone. Withdrawal of that tone is the main neural dilator mechanism, because systemic arterioles have no parasympathetic supply.

· Hormonal constrictors: angiotensin II at AT1 and high-dose vasopressin at V1, both Gq coupled.

· Hormonal dilators: epinephrine at beta-2 receptors, which are Gs coupled and raise cAMP, dilating muscle and liver arterioles at physiologic levels, while at high levels it constricts through alpha-1; natriuretic peptides raise cGMP; histamine and bradykinin dilate through nitric oxide and also open venular gaps, which causes inflammatory edema.

· Metabolic and myogenic control together give autoregulation over a mean arterial pressure of roughly 60 to 70 up to 150 to 160 mm Hg in brain and kidney, and the kidney adds tubuloglomerular feedback.

· Chronic hypertension shifts the autoregulatory range upward, so a fall in pressure that a normotensive person tolerates can cause hypoperfusion.

· Tone itself is sustained partial contraction of the arteriolar smooth muscle, which is what makes both dilation and further constriction possible from the resting state.

Distinctions that matter

· Metabolic control raises flow when the tissue needs more, while myogenic control defends flow against a change in pressure, so they answer different questions even though both act on the same smooth muscle.

· Epinephrine's effect depends on dose and on which receptor predominates in the bed, so the same hormone dilates muscle arterioles at physiologic levels and constricts at high levels.

Common misconceptions

· Expecting parasympathetic fibers to dilate systemic arterioles. They do not supply them, and withdrawal of sympathetic tone is the neural dilator.

· Reading a constrictor response to acetylcholine as an error. It is what happens when the endothelium is damaged and the nitric oxide route is lost.

Where it is taught: Section C06, first three paragraphs and the nitric oxide flow figure of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-BD2

Brain dump 2 · integration and application: Which control wins in which bed

A competent response contains

· Coronary: dominant control is metabolites, namely adenosine, nitric oxide and low O2; hypoxia dilates; the alpha-1 effect is overridden by metabolism.

· Cerebral: dominant control is PaCO2 acting through pH, plus autoregulation; hypoxia dilates below a PaO2 of about 50 mm Hg; sympathetic effect is minor.

· Skeletal muscle: sympathetic at rest and metabolites in exercise; hypoxia dilates; sympathetic effect is strong at rest and overridden in active muscle.

· Skin: sympathetic thermoregulation; hypoxia has little role; sympathetic effect is strong, especially at the arteriovenous anastomoses.

· Pulmonary: hypoxic vasoconstriction, with recruitment and distension as output rises; hypoxia constricts; sympathetic effect is minor.

· Kidney: myogenic control and tubuloglomerular feedback; hypoxia has a minor role; strong sympathetic stimulation constricts.

· The heart extracts about 70 to 75% of arterial oxygen at rest, so there is little extraction reserve and more demand must be met by more flow; metabolic dilation therefore overrides alpha-1 constriction in the coronary bed.

· Hyperventilation lowers PaCO2; CO2 crosses the blood-brain barrier and its removal raises perivascular pH, which constricts cerebral arterioles, so cerebral flow falls despite a normal arterial pressure. Plasma H+ crosses slowly, which is why the CO2 level rather than the blood pH drives the response. Cerebral flow roughly doubles as PaCO2 rises from 40 to 80 mm Hg.

· In exercising muscle, local metabolites and nitric oxide override sympathetic constriction, which is functional sympatholysis, and flow rises about 20-fold.

· After hemorrhage, skin flow serves heat loss rather than metabolism, so strong sympathetic constriction makes the skin pale and cool, while the coronary and cerebral beds are governed locally and escape that constriction.

· Low alveolar PO2 inhibits oxygen-sensitive K+ channels in pulmonary arterial smooth muscle, which depolarizes the cell and opens L-type Ca2+ channels, so the vessel constricts and blood is diverted toward better ventilated alveoli. When the hypoxia involves the whole lung the response is global and raises pulmonary vascular resistance, giving pulmonary hypertension.

· A long-standing hypertensive has an autoregulatory range shifted upward, so a mean pressure that sits comfortably inside a normotensive person's range can fall below the hypertensive person's lower limit, and cerebral and renal flow fall.

Distinctions that matter

· Hypoxia dilates every systemic bed and constricts the pulmonary bed, because the systemic response matches flow to tissue demand while the pulmonary response matches perfusion to ventilation.

· Beds governed locally, the heart and brain, escape sympathetic constriction, while skin, gut, kidney and resting muscle do not, which is what lets the body defend pressure by redistributing flow.

Common misconceptions

· Expecting the heart to meet extra demand by extracting more oxygen. Extraction is already about 70 to 75% at rest, so flow has to rise.

· Explaining hyperventilation lightheadedness by low oxygen. The PaO2 is high; it is the low PaCO2 and the resulting alkaline perivascular pH that constrict cerebral arterioles.

Where it is taught: Section C06, the paragraph on dominant control in each bed, the table of beds and the stop and predict box of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-RT01

Retrieval exercise: Rebuild the nitric oxide pathway

A competent response contains

· Step 1, endothelium: shear stress, or acetylcholine and bradykinin, act on the endothelial cell.

· Step 2, endothelium: Ca2+-calmodulin and phosphorylation activate eNOS.

· Step 3, endothelium: eNOS converts L-arginine to nitric oxide and citrulline.

· Step 4, smooth muscle: nitric oxide diffuses across and activates soluble guanylyl cyclase.

· Step 5, smooth muscle: cGMP rises and activates protein kinase G.

· Step 6, smooth muscle: cytosolic Ca2+ falls and myosin light chain phosphatase is activated.

· Step 7, smooth muscle: the myosin light chains are dephosphorylated and the arteriole dilates.

· The enzyme in the endothelium is eNOS, its substrate is L-arginine, and its products are nitric oxide and citrulline.

· In the smooth muscle, nitric oxide activates soluble guanylyl cyclase, which makes cGMP, which activates protein kinase G.

· Nitrate drugs release nitric oxide inside the smooth muscle cell, so they enter the pathway at step 4 and bypass eNOS, which is why they still work when the endothelium is dysfunctional.

· The dilation spreads upstream to feeding arteries, which is flow-mediated dilation.

Distinctions that matter

· The pathway crosses two cells: everything up to the production of nitric oxide happens in the endothelium, and everything after it happens in the smooth muscle.

· Nitric oxide lowers Ca2+ and also activates myosin light chain phosphatase, so it reduces myosin light chain phosphorylation from both directions.

Common misconceptions

· Placing eNOS in the smooth muscle. It is endothelial, which is why endothelial damage removes the physiologic route to dilation while nitrates still work.

· Treating cGMP as acting directly on the contractile proteins. It works through protein kinase G.

Where it is taught: Section C06, the paragraph on endothelial control and the nitric oxide flow figure of the CV-11 Hemodynamics and Blood Flow Regulation notes.

NPLEX-P1-CARD-11-01-C06-Q02

Board-style question: Regional flow during hyperventilation

Correct answer: A. A rise in perivascular pH that constricts cerebral arterioles and reduces cerebral blood flow

Why it is right: Removing CO2 raises perivascular pH, and cerebral arterioles constrict as pH rises, so cerebral blood flow falls even though mean arterial pressure sits well inside the autoregulatory range. This is the same relationship that roughly doubles cerebral flow as PaCO2 rises from 40 to 80 mm Hg, running in the opposite direction.

B is wrong: Mean arterial pressure here is about 93 mm Hg, comfortably above the lower autoregulatory limit of roughly 60 to 70 mm Hg, so autoregulatory failure does not apply.

C is wrong: Inhibition of oxygen-sensitive potassium channels is the mechanism of hypoxic vasoconstriction in pulmonary arterial smooth muscle, and in any case the PaO2 is 108 mm Hg.

D is wrong: Loss of myogenic tone with a rise in cerebral blood flow is the opposite of what a low PaCO2 produces.

A competent response contains

· Hyperventilation lowers PaCO2 to 22 mm Hg, CO2 leaves the brain across the blood-brain barrier, and perivascular pH rises.

· An alkaline perivascular pH constricts cerebral arterioles, so cerebral blood flow falls while arterial pressure stays normal.

Distinctions that matter

· Cerebral flow follows PaCO2 rather than arterial pH, because CO2 crosses the blood-brain barrier readily and plasma H+ crosses slowly.

Common misconceptions

· Looking for a fall in arterial pressure to explain the symptom. The pressure is 128/76 mm Hg and the problem is local arteriolar constriction.

Where it is taught: Section C06 of the CV-11 Hemodynamics and Blood Flow Regulation notes.